# Class 1

Introduction

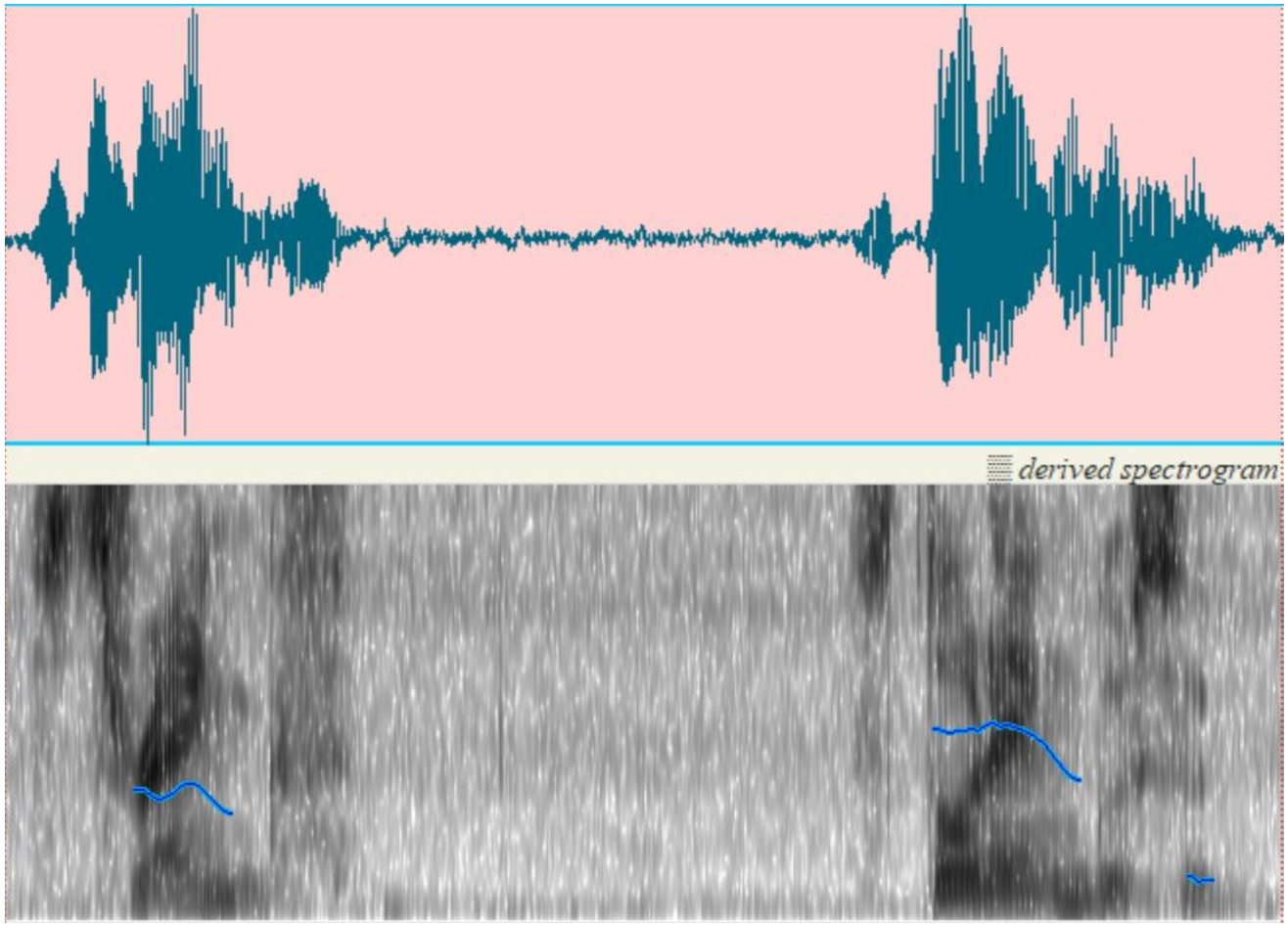
# Phonetics

# A brief history…

- International Phonetic Association – founded by Paul Passy [pasi] and colleagues in 1886
- The International Phonetic Alphabet (IPA) was first published in 1888
  - Most recent revision: 2005
  - Currently has 83 consonants, 28 vowels
- **Goals of IPA**
  - Be universal
  - Be unambiguous

# Visualizing your sounds

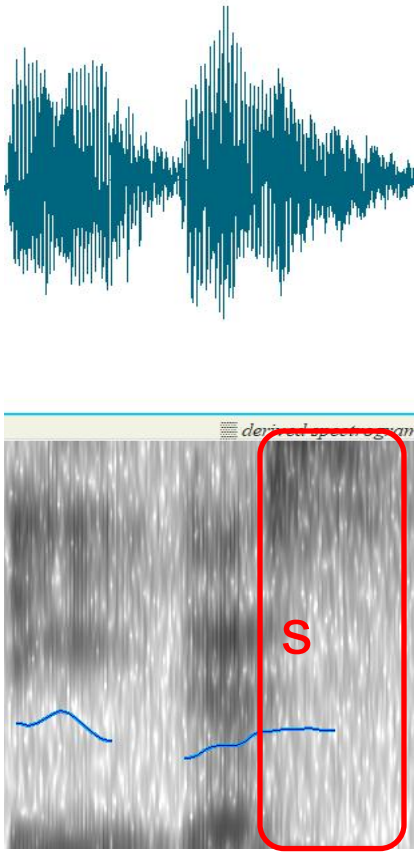
- Strength
- ストレングス



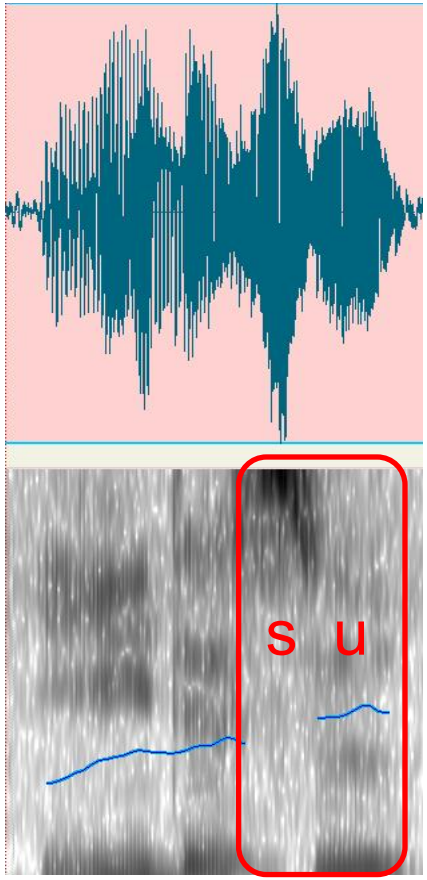
# Visualizing your sounds

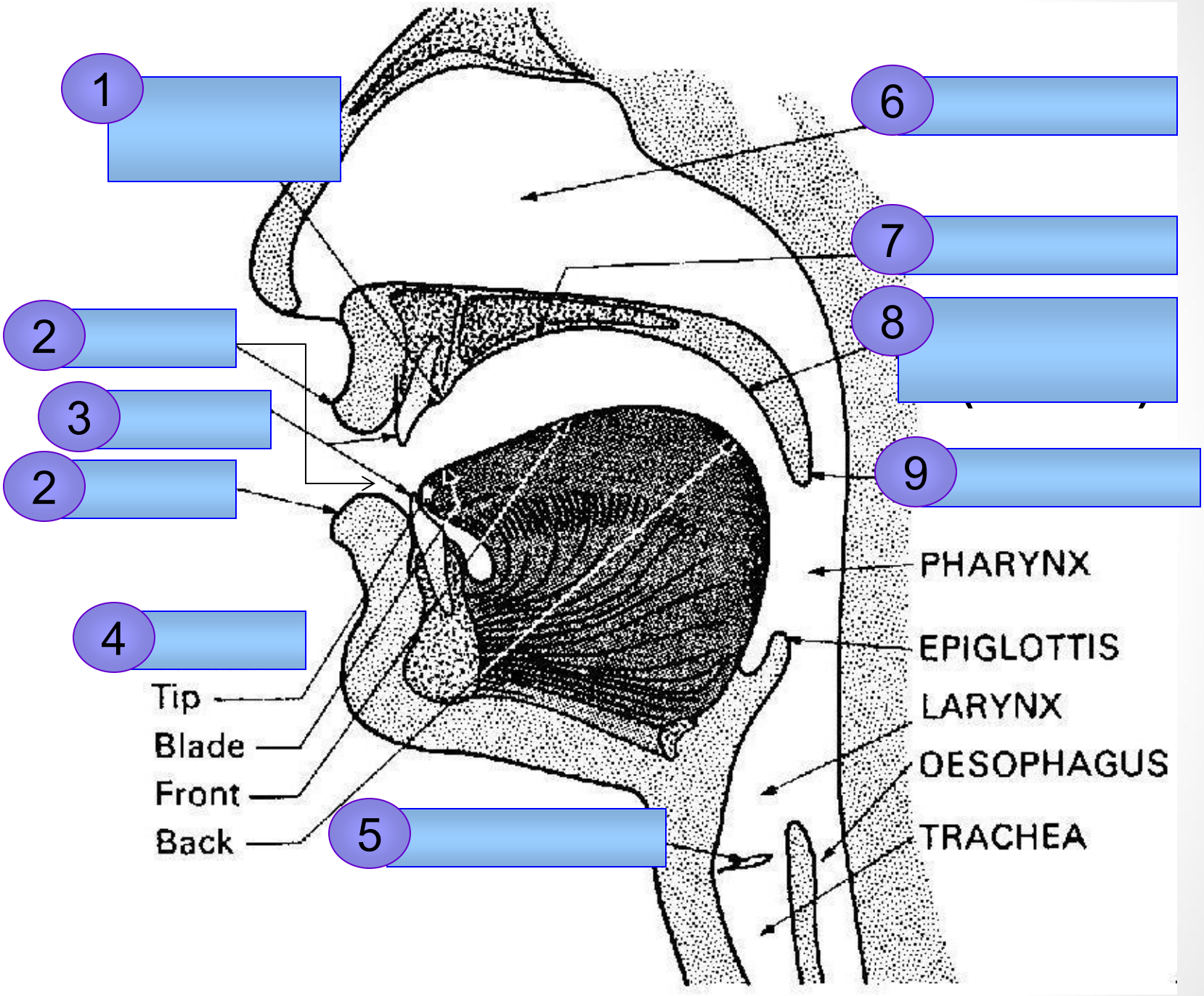
- 「いいです」 'Good'

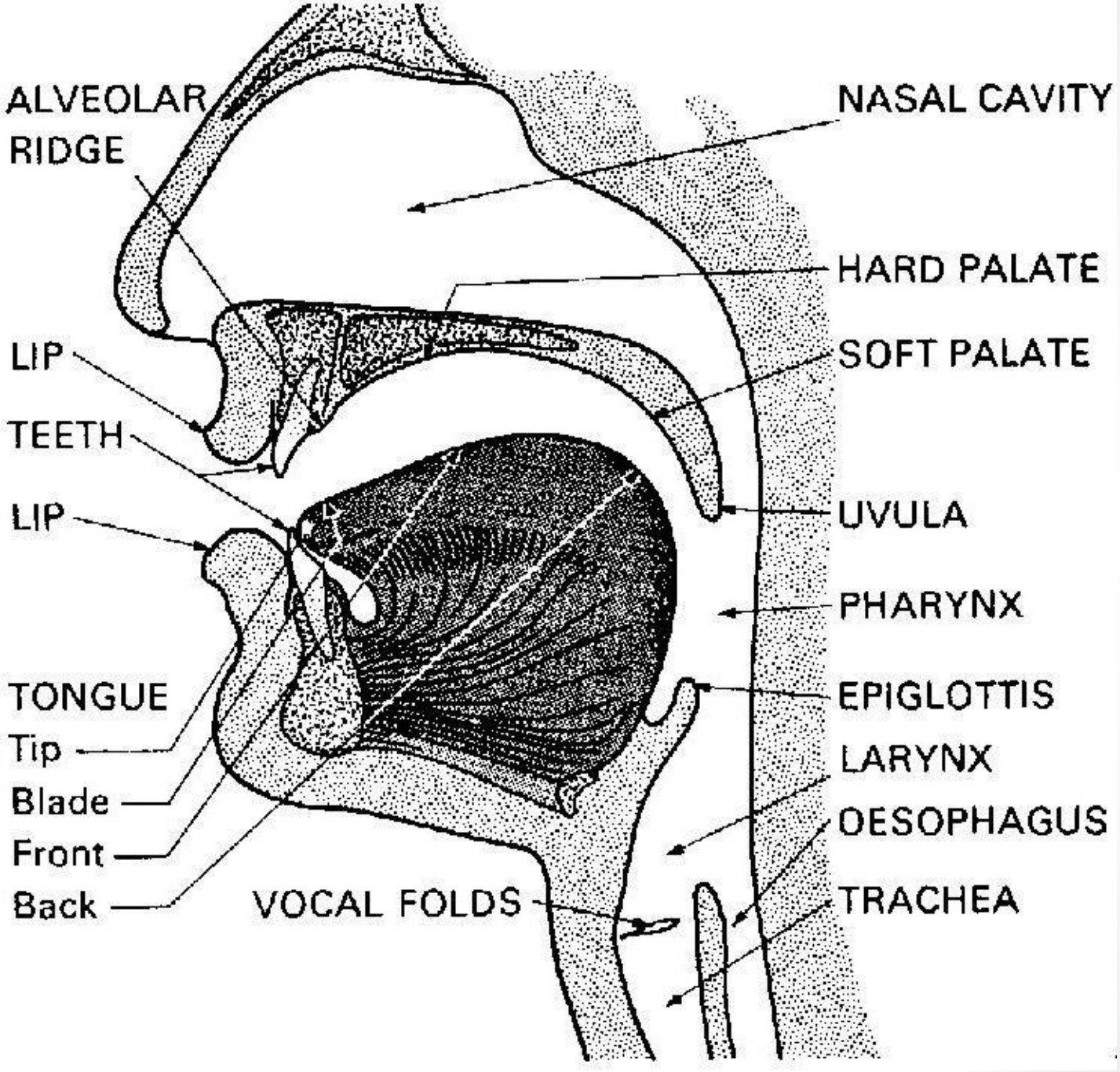
Kanto dialects



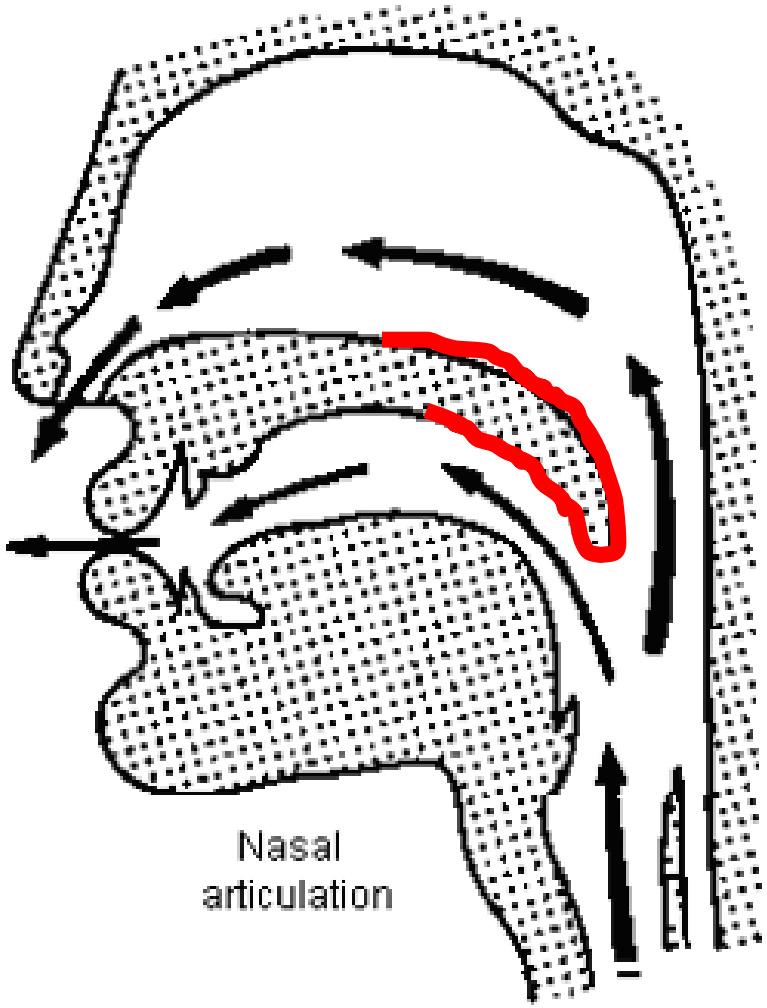

Kyoto dialects

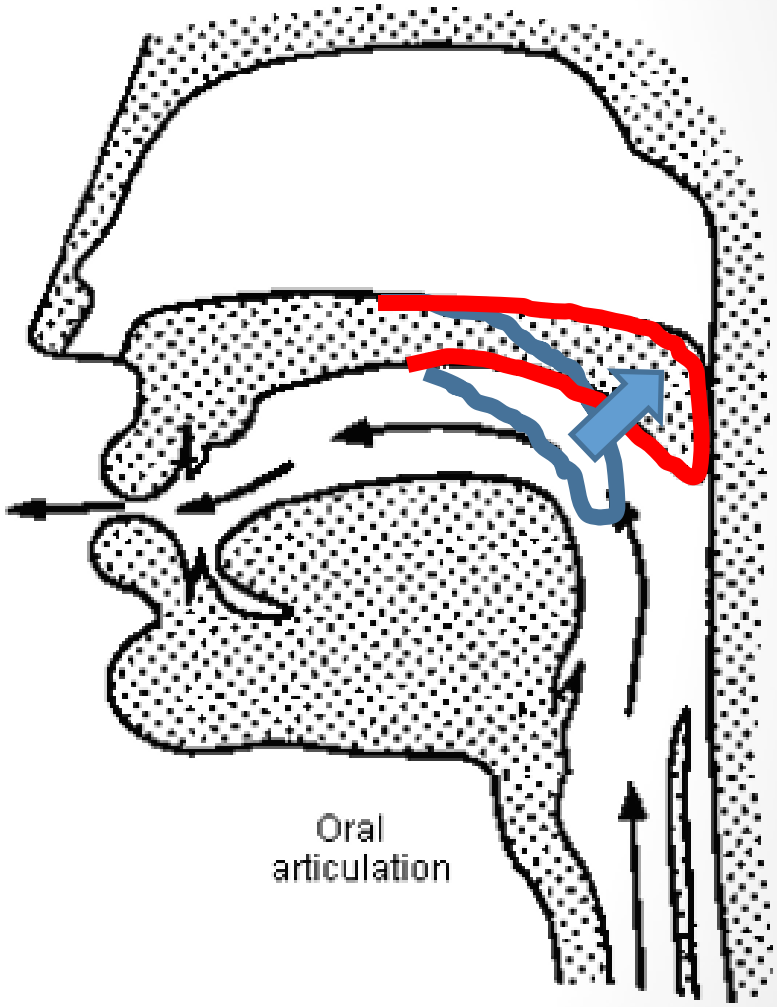



1
6
7
2
8
3
2
9
PHARYNX
4
EPIGLOTTIS
Tip
LARYNX
Blade
OESOPHAGUS
Front
5
TRACHEA
Back


ALVEOLAR RIDGE
NASAL CAVITY
HARD PALATE
SOFT PALATE
LIP
TEETH
LIP
UVULA
PHARYNX
TONGUE
Tip
Blade
Front
Back
EPIGLOTTIS
LARYNX
OESOPHAGUS
VOCAL FOLDS
TRACHEA

# Velum



# Class 2

Consonants

# PARAMETERS FOR DESCRIBING CONSONANTS

# 1 Voicing and aspiration

- **1 Voicing and aspiration** (= what the larynx is doing)
  - Voiced sound = vocal folds vibrating (e.g., /z/)
  - Voiceless sound = vocal folds **not** vibrating (e.g., /s/)
  - Aspiration = vocal folds held apart long enough to allow an extra puff of air to exit the mouth
- **2 Place of articulation** (which articulators are moving, and where)
- **3 Manner of articulation** (what sort of constriction and how much)

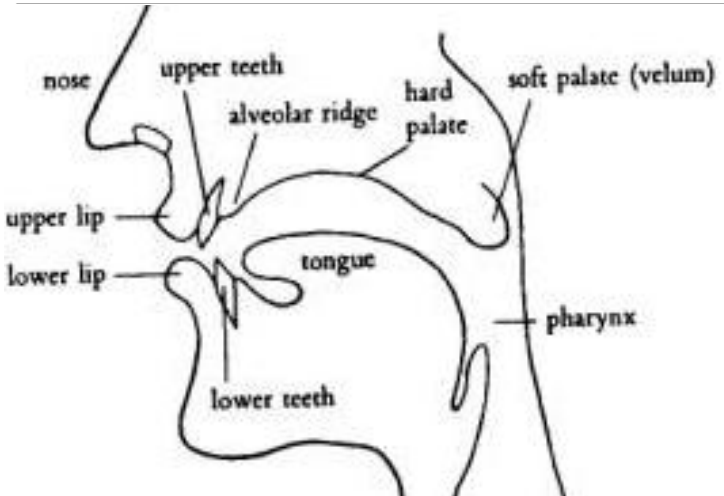
# 2&3 Place of articulation

- **1 Voicing and aspiration** (= what the larynx is doing)
  - Voiced sound = vocal folds vibrating (e.g., /z/)
  - Voiceless sound = vocal folds **not** vibrating (e.g., /s/)
  - Aspiration = vocal folds held apart long enough to allow an extra puff of air to exit the mouth
- **2 Place of articulation** (which articulators are moving, and where)
- **3 Manner of articulation** (what sort of constriction and how much)

# Manner and Place of articulation

CONSONANTS (PULMONIC) © 2005 IPA

| | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Plosive | p b | | | t d | | ʈ ɖ | c ɟ | k g | q ɢ | | ʔ |
| Nasal | m | ɱ | | n | | ɳ | ɲ | ŋ | ɴ | | |
| Trill | ʙ | | | r | | | | | ʀ | | |
| Tap or Flap | | ⱱ | | ɾ | | ɽ | | | | | |
| Fricative | ɸ β | f v | θ ð | s z | ʃ ʒ | ʂ ʐ | ç ʝ | x ɣ | χ ʁ | ħ ʕ | h ɦ |
| Lateral fricative | | | | ɬ ɮ | | | | | | | |
| Approximant | | ʋ | | ɹ | | ɻ | j | ɰ | | | |
| Lateral approximant | | | | l | | ɭ | ʎ | ʟ | | | |




| Labial | Coronal | Dorsal | Radical | Laryngeal |
|---|---|---|---|---|
| bilabial | (inter)dental | velar | Pharyngeal | Glottal |
| labiodental | alveolar | uvular | | |
| | postalveolar | | | |
| | retroflex | | | |
| | palatal | | | |

# 3 Manner of articulation

- **Oral stop/plosive**: complete closure, pressure builds up
  - Active and passive articulators are brought together to make a complete closure so that airflow out of the mouth is cut off

[p]

- **Nasal stop**: open velum, resonance in nasal cavity
  - The velum is open and air flows freely out the nose, but there's complete closure somewhere else

[m]

- **Fricative**: incomplete closure, turbulent air
  - Articulators are brought close together but not closed completely; the stream of air forced between them becomes turbulent and noisy

[v]

- **Affricate**: stop + fricative
  - A sequence of stop plus fricative combined to form a single sound

[dʒ]

- **Approximant**: narrowed vocal tract, but no frication
  - The active articulator moves to narrow the vocal tract, but not so much that fricative noise is created
  - Subtypes:
    - glides
    - liquids (laterals & rhotics)

[j]

# 3 Manner of articulation

## *Approximants*

- **Glides** - [w] ‘well’ and [j] ‘you’
  - Vowel-like quality
- **Laterals** - [l]
  - Air flows over sides of tongue
  - American [l] is alveolar; what about laterals in other languages (e.g. Spanish, Italian?)
- **Rhotics** - [ɹ]
  - The American English ‘r’ is upside-down in the IPA because it is a retroflex consonant
  - Rhotics vary widely across world’s languages - examples?

# Manner and Place of articulation

CONSONANTS (PULMONIC) © 2005 IPA

| | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Plosive | p b | | t d | | | ʈ ɖ | c ɟ | k g | q ɢ | | ʔ |
| Nasal | m | ɱ | n | | | ɳ | ɲ | ŋ | ɴ | | |
| Trill | ʙ | | r | | | | | | ʀ | | |
| Tap or Flap | | ⱱ | ɾ | | | ɽ | | | | | |
| Fricative | ɸ β | f v | θ ð | s z | ʃ ʒ | ʂ ʐ | ç ʝ | x ɣ | χ ʁ | ħ ʕ | h ɦ |
| Lateral fricative | | | | ɬ ɮ | | | | | | | |
| Approximant | | ʋ | ɹ | | | ɻ | j | ɰ | | | |
| Lateral approximant | | | l | | | ɭ | ʎ | ʟ | | | |

**We classify consonants by place and manner of articulation**

(a) voiced or voiceless?

(b) place? (bilabial, alveolar, etc.)

(c) manner? (stop, fricative, affricate, etc.)

# EXERCISES

# Consonant Practice

**Exercise 1: Make the sound**

- (a) Voiced alveolar (oral) stop
- (b) Voiceless labiodental (oral) fricative
- (c) Voiced bilabial nasal stop

CONSONANTS (PULMONIC) © 2005 IPA

| | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Plosive | p b | | | t d | | ʈ ɖ | c ɟ | k g | q ɢ | | ʔ |
| Nasal | m | ɱ | | n | | ɳ | ɲ | ŋ | ɴ | | |
| Trill | ʙ | | | r | | | | | ʀ | | |
| Tap or Flap | | ⱱ | | ɾ | | ɽ | | | | | |
| Fricative | ɸ β | f v | θ ð | s z | ʃ ʒ | ʂ ʐ | ç ʝ | x ɣ | χ ʁ | ħ ʕ | h ɦ |
| Lateral fricative | | | | ɬ ɮ | | | | | | | |
| Approximant | | ʋ | | ɹ | | ɻ | j | ɰ | | | |
| Lateral approximant | | | | l | | ɭ | ʎ | ʟ | | | |

# Consonant Practice

**Exercise 2: Make the sound and describe the parameters:**

- (a) [p]
- (b) [ɹ]
- (c) [g]

CONSONANTS (PULMONIC) © 2005 IPA

| | Bilabial | Labiodental | Dental | Alveolar | Postalveolar | Retroflex | Palatal | Velar | Uvular | Pharyngeal | Glottal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Plosive | p b | | | t d | | ʈ ɖ | c ɟ | k g | q ɢ | | ʔ |
| Nasal | m | ɱ | | n | | ɳ | ɲ | ŋ | ɴ | | |
| Trill | ʙ | | | r | | | | | ʀ | | |
| Tap or Flap | | ⱱ | | ɾ | | ɽ | | | | | |
| Fricative | ɸ β | f v | θ ð | s z | ʃ ʒ | ʂ ʐ | ç ʝ | x ɣ | χ ʁ | ħ ʕ | h ɦ |
| Lateral fricative | | | | ɬ ɮ | | | | | | | |
| Approximant | | ʋ | | ɹ | | ɻ | j | ɰ | | | |
| Lateral approximant | | | | l | | ɭ | ʎ | ʟ | | | |

## Exercise 3:

- voiced velar nasal stop [ ŋ ]

  _____

- voiceless oral glottal fricative [ h ]

  _____

- voiceless oral alveopalatal affricate [ ʧ ]

  _____

# Exercise 4:

## Guess the Sound!

1. Place

Bilabial

2. Manner

Plosive

3. Voicing

Voiceless

IPA CONSONANTS 1

# Exercise 4:

## Guess the Sound!

1. Place

Bilabial

2. Manner

Plosive

3. Voicing

Voiceless

IPA CONSONANTS 1

# Exercise 4:

## Guess the Sound!

1. Place

Bilabial

2. Manner

Plosive

3. Voicing

Voiceless

IPA CONSONANTS 1

[ p ]

CONSONANT CHART

| | Bilab. | L-D. | Dent. | Alv. | P-A. | Pal. | Vel. | Glo. |
|---|---|---|---|---|---|---|---|---|
| Plosive | p b | | | t d | | | k g | ʔ |
| Nasal | m | | | n | | | ŋ | |
| Tap/Flap | | | | ɾ | | | | |
| Fricat. | | f v | θ ð | s z | ʃ ʒ | | | h |
| Approx. | w | | | ɹ | | j | | |
| Lat. Appr. | | | | l | | | | |

# IPA 10 questions

**Goal:** Guess the hidden sound using only Yes/No questions!

**How to Play:**

1. **Draw** one card. Keep it a secret!
2. **Ask** "Yes/No" questions to guess the card.
3. **Think** about Place, Manner, and Voicing.
4. Whoever shouts the correct IPA sound first gets the card!

🗣️ **Useful Phrases:**

YES:

"Is it voiced / voiceless?"
"Is it an alveolar sound?"

NO: "Tell us about the manner of articulation of that sound."

# IPA Karuta

**Goal: Be the first to grab the correct sound!**

**How to Play:**

1. **Spread** the cards face-up on your desk.
2. **Hands on your head!** (Wait for the teacher).
3. **Listen** carefully to the sound or word.
4. **Slap** the card with the matching features!

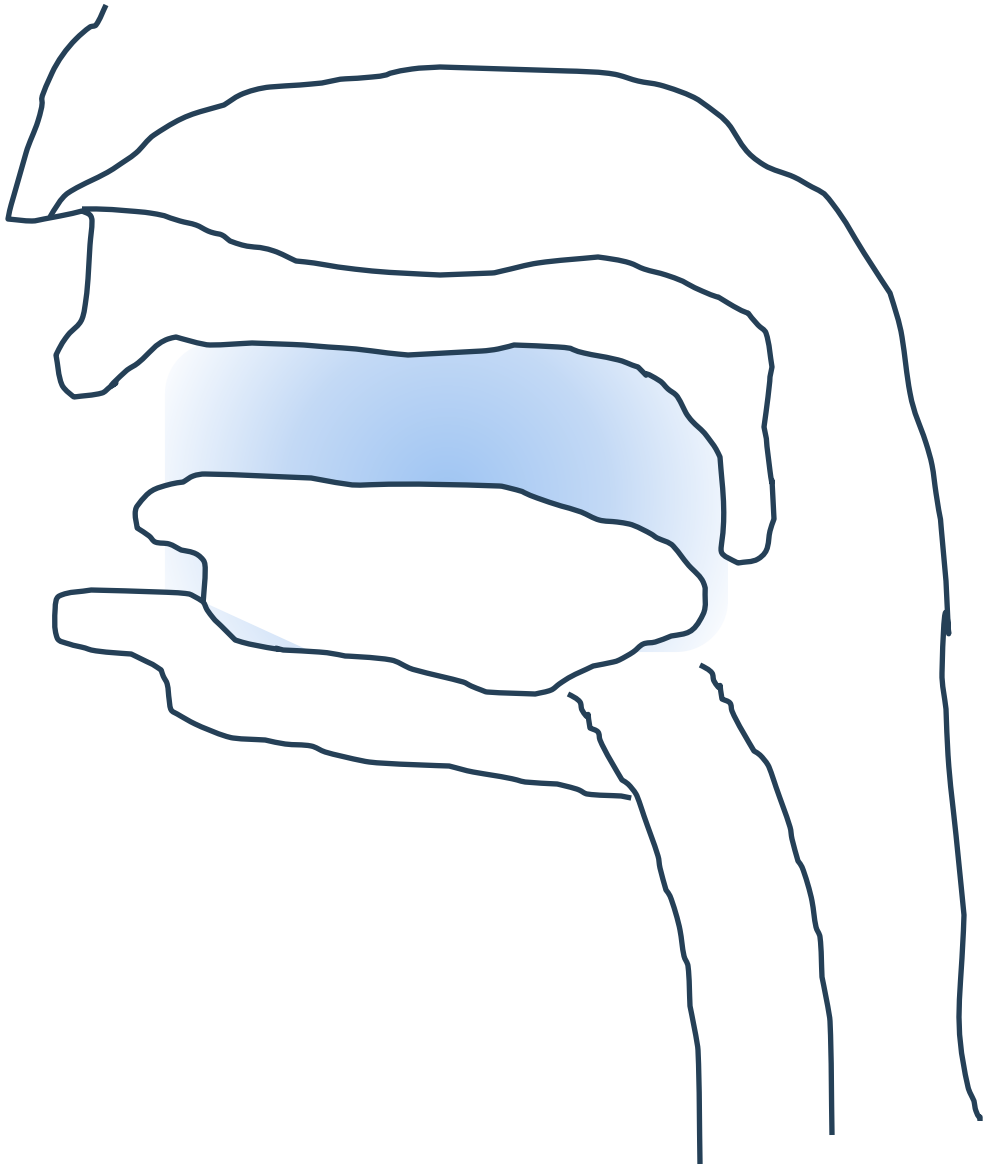
   *(e.g., Hear /m/ ➔ Grab "Bilabial / Nasal / Voiced")*

⚠️ **Rule:**
False start? You must sit out for one turn!

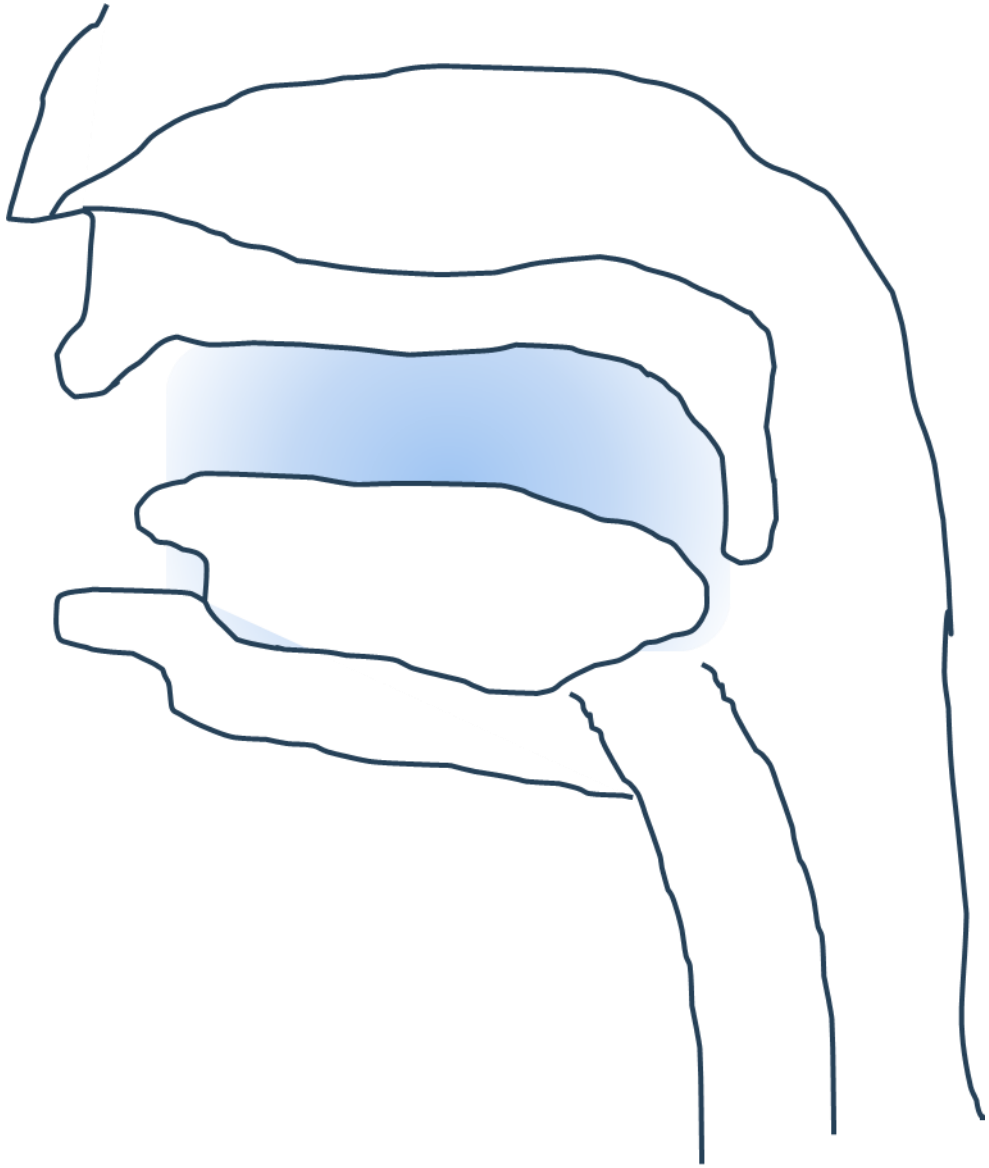
# Class 3

Vowels

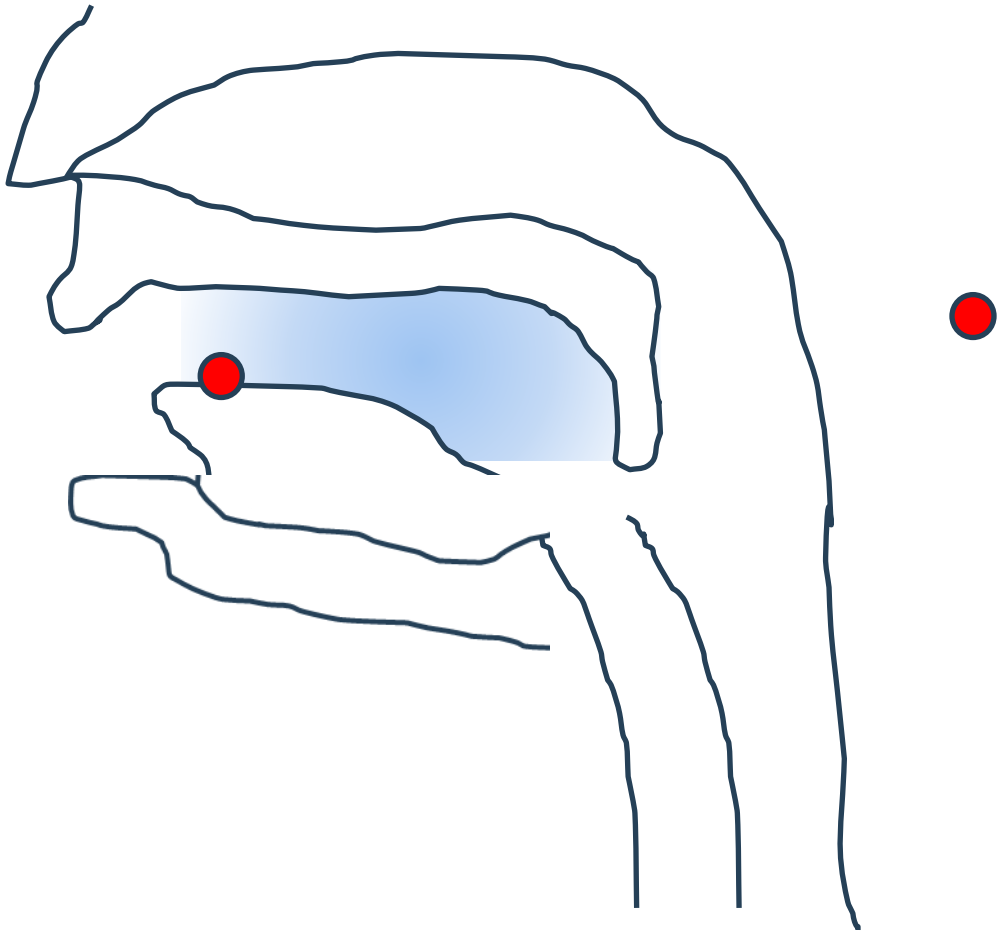
# It is the lower jar that is moving.



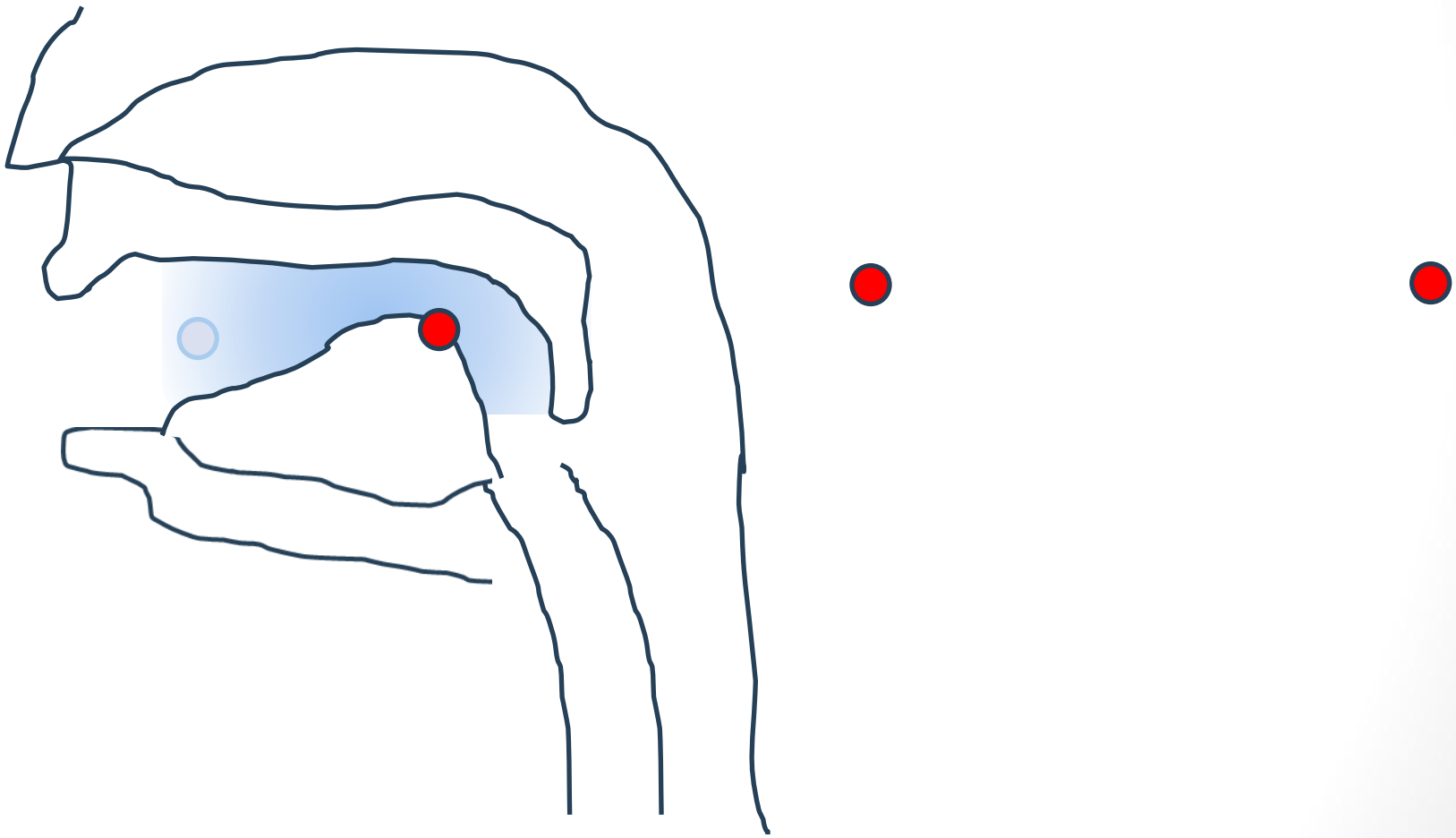
# It is the lower jar that is moving.



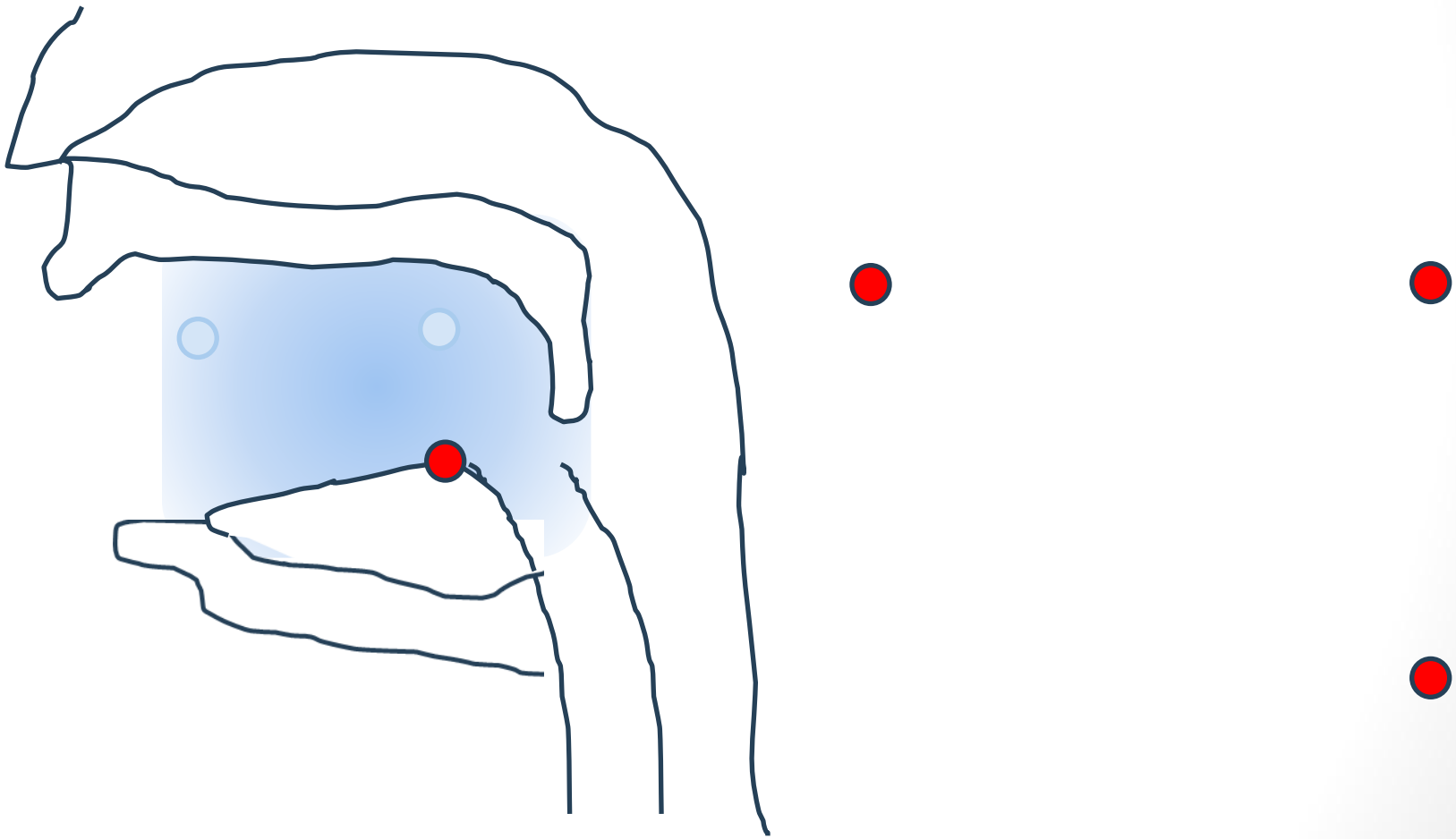
# Vowel gird



# Vowel gird



# Vowel gird



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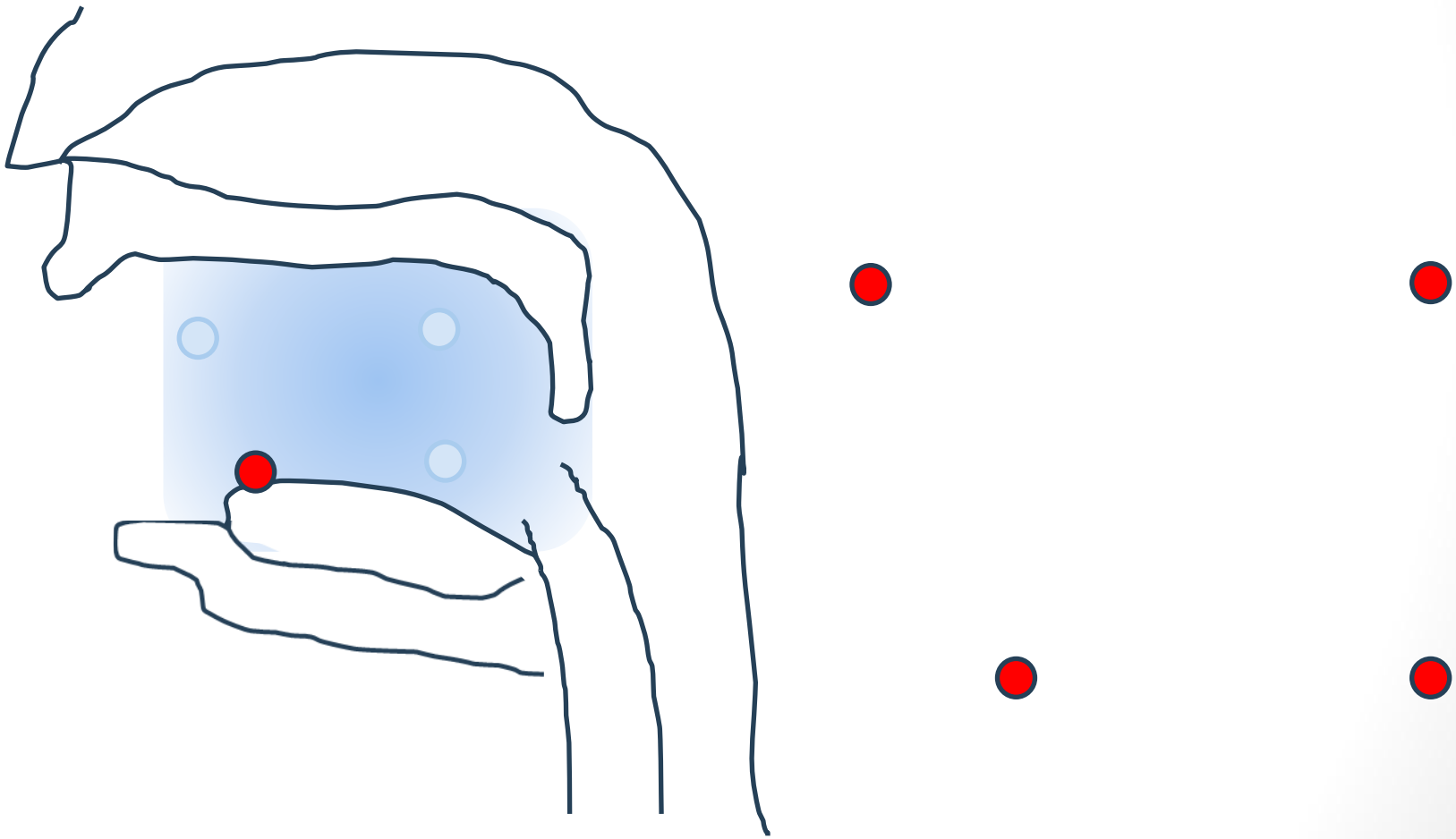

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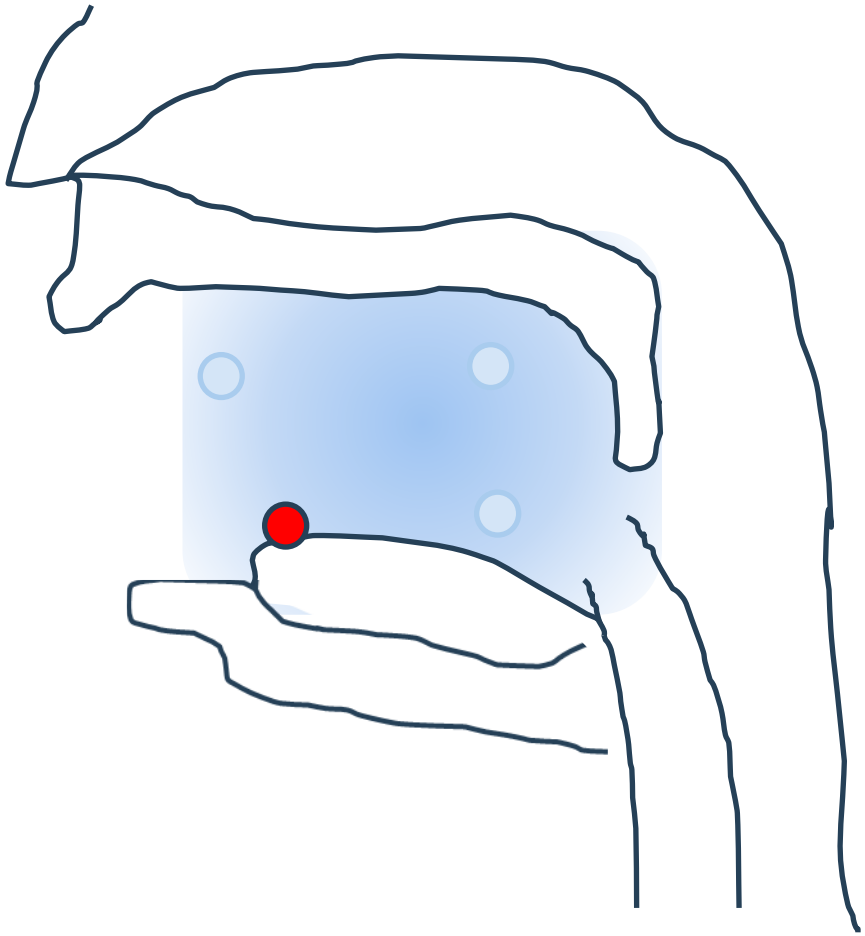
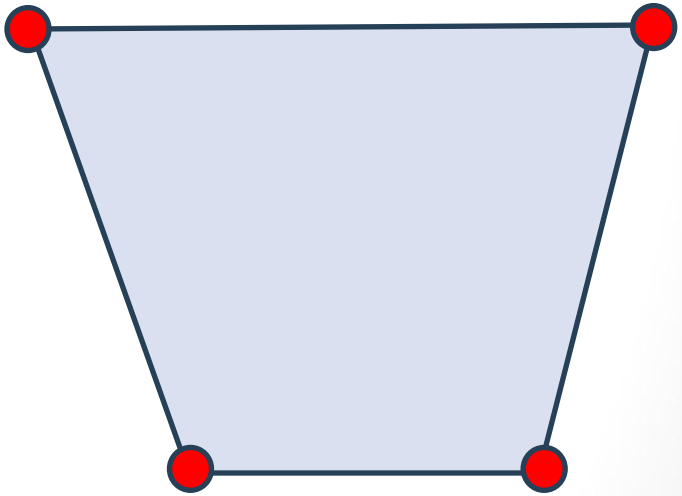
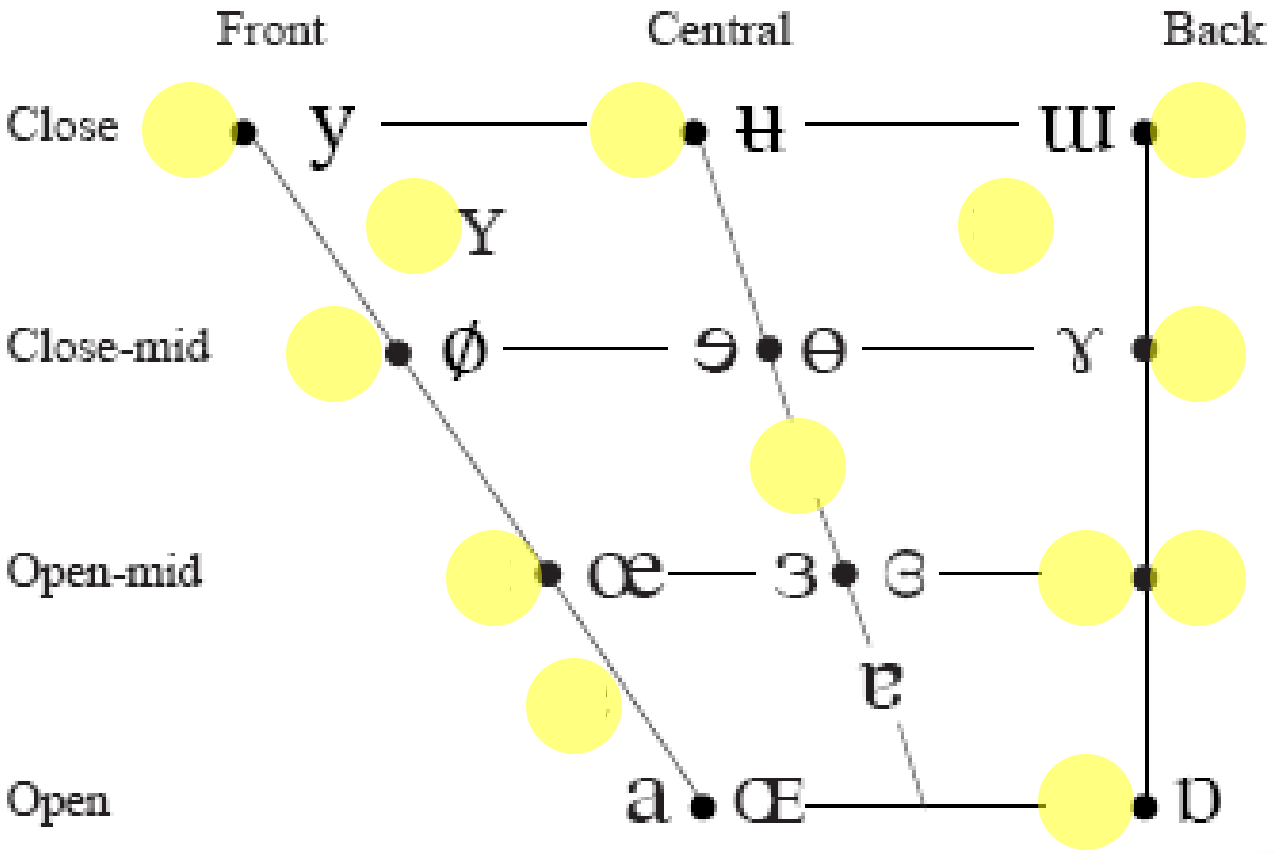




# Which vowel symbols go where?



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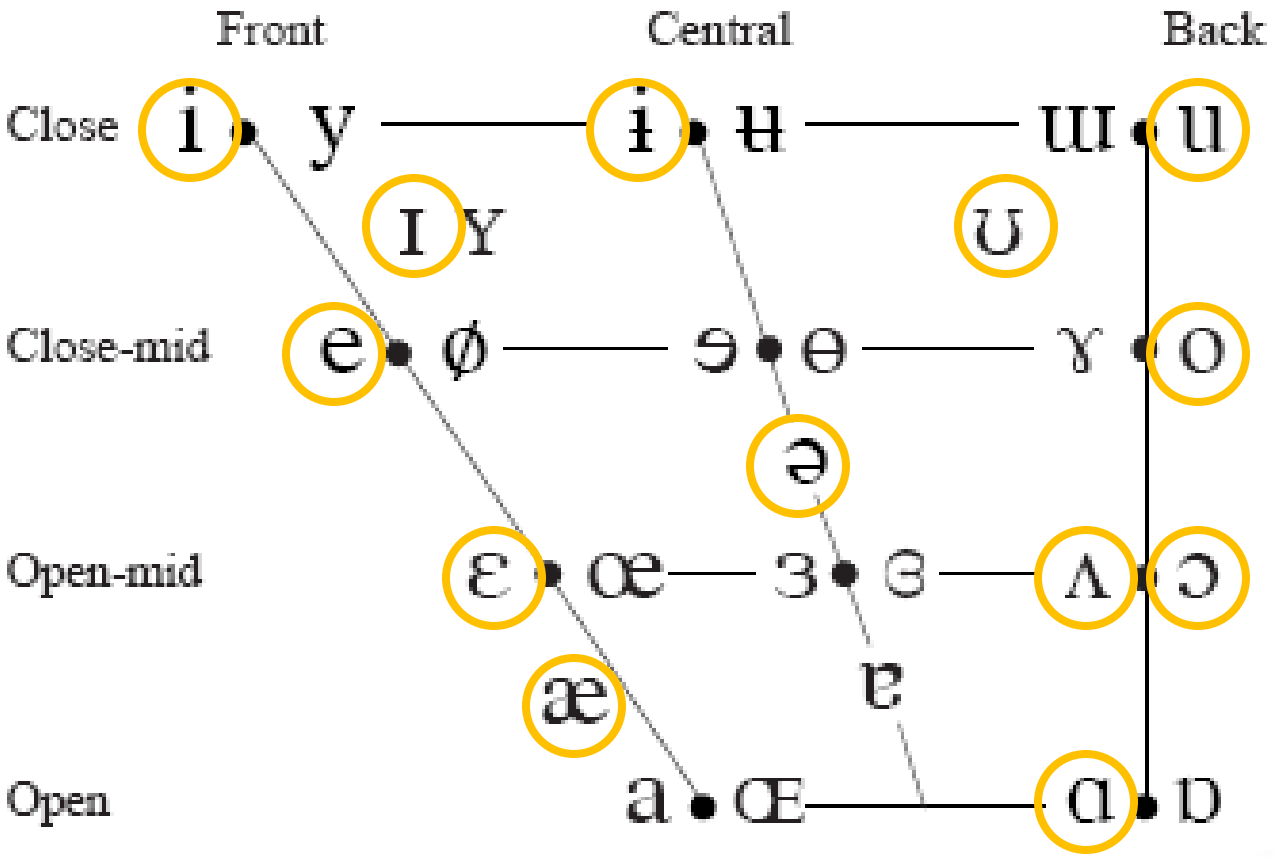

# IPA 10 questions

**Goal:** Guess the hidden sound using only Yes/No questions!

**How to Play:**

1. **Draw** one card. Keep it a secret!
2. **Ask** "Yes/No" questions to guess the card.
3. **Think** about Frontness, Height, Roundedness, and Tenseness.
4. Whoever shouts the correct IPA sound first gets the card!

🗣 **Useful Phrases:**

YES:

"Is it voiced / voiceless?"
"Is it an alveolar sound?"

# IPA Karuta

**Goal: Be the first to grab the correct sound!**

**How to Play:**

1. **Spread** the cards face-up on your desk.
2. **Hands on your head!** (Wait for the teacher).
3. **Listen** carefully to the sound or word.
4. **Slap** the card with the matching features!

   *(e.g., Hear* [ i ] ➔ *Grab "Close / Front / Unrounded / Tense")*

⚠️ **Rule:**
False start? You must sit out for one turn!